

QUBIT ELECTRONICS — FORMULA SHEET

Martina Paola Calvi

1. QUBIT FUNDAMENTALS

State & Normalisation

- General single-qubit state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
- Normalisation: $|\alpha|^2 + |\beta|^2 = 1 = \langle\psi|\psi\rangle$
- Conjugate (bra): $\langle\psi| = |\psi\rangle^\dagger = [\alpha^*, \beta^*]$
- n -qubit state ($n = 3$ example): $|\psi\rangle = \sum_{k=0}^7 c_k |k\rangle$, $\sum |c_k|^2 = 1$

Bloch Sphere

- $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$
- Bloch vector: $\vec{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)^T$
- Poles / equator: $N = (0, 0, 1)$, $S = (0, 0, -1)$, $\pm x = (\pm 1, 0, 0)$, $\pm y = (0, \pm 1, 0)$

Born Rule & Measurement

- Probability of outcome $|a\rangle$: $P(|a\rangle) = |\langle a|\psi\rangle|^2$
- Expectation: $\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle = \text{Tr}(\rho\hat{O})$

Standard Basis States

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle), \quad |\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$$

Single-Qubit Gates (Paulis & Clifford)

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Rotation gates ($\hat{n} \in \{X, Y, Z\}$):

$$R_{\hat{n}}(\theta) = e^{-i\theta\hat{n}/2} = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}\hat{n}$$

Two-Qubit Gate: CNOT

- Action: $|A\rangle|B\rangle \rightarrow |A\rangle|A \oplus B\rangle$

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Density Matrix

- Pure state: $\rho = |\psi\rangle\langle\psi|$; mixed: $\hat{\rho} = \sum_i p_i |\Psi_i\rangle\langle\Psi_i|$
- Purity: $\text{Tr}(\rho^2) = 1$ (pure); $\text{Tr}(\rho^2) < 1$ (mixed)
- Bloch representation: $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$
- Expectation: $\langle\hat{O}\rangle = \text{Tr}(\rho\hat{O})$

Coherence Times & Gate Metrics

- $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}$
- $\frac{1}{T_2^*} = \frac{1}{2T_1} + \frac{1}{T_\phi}$ (inhomogeneous)
- Max coherent gates: $N_{\text{gates}} \sim T_2/t_{\text{gate}}$
- Relaxation rates: $\Gamma_1 = 1/T_1$; $\Gamma_\phi = 1/T_\phi$; $\Gamma_2 = \Gamma_1/2 + \Gamma_\phi$
- Thermal equilibrium: $\Gamma_{1\uparrow} = \Gamma_{1\downarrow} e^{-\hbar\omega/k_B T}$

2. SUPERCONDUCTING QUBITS

Cooper Pairs & Superconductivity

- Macroscopic wavefunction: $\psi = \psi_0 \exp[-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{x})]$
- Probability density: $\rho = |\psi_0|^2$
- Meissner effect: $B = 0$ inside superconductor

Josephson Junction

- Phase difference: $\phi = \varphi_2 - \varphi_1$
- DC Josephson: $I = I_C \sin\phi$
- AC Josephson: $V = \frac{\Phi_0}{2\pi} \dot{\phi}$, $\Phi_0 = \frac{h}{2e} = \frac{2\pi\hbar}{2e}$
- Josephson inductance: $L_J = \frac{\Phi_0}{2\pi I_C}$, $L(\phi) = \frac{L_J}{\cos\phi}$
- Josephson energy: $E_J(\phi) = E_J(1 - \cos\phi)$, $E_J = \frac{\Phi_0 I_C}{2\pi}$
- Charging energy: $E_C = \frac{e^2}{2C}$

LC Resonator Quantisation

- Hamiltonian: $H = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L} = \hbar\omega_r(\hat{a}^\dagger\hat{a} + \frac{1}{2})$
- Resonance: $\omega_{01} = 1/\sqrt{LC}$
- Canonical commutation: $[\hat{\Phi}, \hat{Q}] = i\hbar$; $[\hat{a}, \hat{a}^\dagger] = 1$
- Zero-point fluctuations: $Q_{\text{zpf}} = \sqrt{\hbar/(2Z_0)}$; $\Phi_{\text{zpf}} = \sqrt{\hbar Z_0/2}$

Transmon Qubit

- Regime: $E_J \gg E_C$ (charge-noise insensitive)
- Transition frequency: $\hbar\omega_{01} \approx \sqrt{8E_J E_C} - E_C \approx 10$ GHz
- Anharmonicity: $\delta = (E_{12} - E_{01})/\hbar < 0$; $E_{01} \neq E_{12}$
- Flux-tunable inductance: $L(\Phi) = \frac{\Phi_0}{2\pi I_C \cos(2\pi\Phi/\Phi_0)}$
- Operating temperature: $k_B T \ll \hbar\omega_{01}$
- π -pulse time: $T_\pi \approx 4$ –20 ns; $T_1, T_2 \sim 100$ –1000 μ s

3. CIRCUIT QED

Light–Matter Coupling

- Dipole interaction: $H = -\vec{d} \cdot \vec{E}(t)$
- Rabi frequency: $\Omega_R = dE_0/\hbar$
- Vacuum electric field (cavity): $E_0 \approx 0.2$ V/m; enhancement $\sim 2 \times 10^4$

Resonator

- Mode condition: $2L = m\lambda_m$; FSR: $\Delta f = c/(2L)$
- Angular frequency: $\omega_r = 1/\sqrt{L_r C_r}$
- Quality factor: $Q = \omega_r/\kappa$; $\kappa = \omega_r/Q$ (photon decay rate)

Jaynes–Cummings Hamiltonian

$$H_{JC} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2} \hat{\sigma}_z + \hbar g(\hat{a} \hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$$

Rotating-wave approximation (RWA) keeps only energy-conserving terms.

Coupling Strength

- $g = \omega_r \frac{C_g}{C} \left(\frac{E_J}{2E_C} \right)^{1/4} \sqrt{\frac{\pi Z_r}{R_k}}$
- Typical values: $g/2\pi \approx 100$ MHz–1 GHz

Dispersive Regime ($|\Delta| \gg g$)

- Detuning: $\Delta = \omega_{01} - \omega_r$
- Dispersive shift: $\chi = g^2/\Delta$
- Effective Hamiltonian:

$$H_{\text{disp}} \approx (\omega_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2} \hat{\sigma}_z$$

- Qubit-state-dependent resonator frequency: $\omega_r^{(\uparrow/\downarrow)} = \omega_r \mp \chi$
- Purcell rate: $\gamma_p = (g/\Delta)^2 \kappa$; optimal: $\chi/\kappa \approx 1$

Full System Hamiltonian

$$H = \frac{Q_r^2}{2C_r} + \frac{\Phi_r^2}{2L_r} + 4E_C \hat{n}^2 - E_J \cos \hat{\phi} - g(\hat{b}^\dagger - \hat{b})(\hat{a}^\dagger - \hat{a})$$

4. QUBIT CONTROL & READOUT

Microwave Drive (Transmon)

- Drive Hamiltonian: $\hat{H} = \frac{\hat{Q}^2}{2C_\Sigma} + \frac{\hat{\Phi}^2}{2L} + \frac{C_d}{C_\Sigma} V_d(t) \hat{Q}$
- Charge operator: $\hat{Q} = -iQ_{\text{zpf}}(\hat{a} - \hat{a}^\dagger)$
- Impedance: $Z_0 = \sqrt{L/C_\Sigma}$
- In qubit subspace ($\hat{a} \rightarrow \hat{\sigma}_-$): $\hat{H} = -\frac{\omega_q}{2} \hat{\sigma}_z + \Omega V_d(t) \hat{\sigma}_y$

Rotating Frame & RWA

- Frame transformation: $U(t) = e^{i\omega_q t \hat{\sigma}_z/2}$
- IQ drive: $V_d(t) = V_0 s(t) [I \sin(\omega_d t) - Q \cos(\omega_d t)]$
- Rotating-frame drive ($\omega_d \approx \omega_q$):

$$\hat{H}'_d = -\frac{\Omega}{2} V_0 s(t) (I \hat{\sigma}_x + Q \hat{\sigma}_y)$$

Virtual Z Gate (Phase Control)

- IQ phase rotation by ϕ :

$$I' = I \cos \phi - Q \sin \phi, \quad Q' = I \sin \phi + Q \cos \phi$$

- $\phi = \pi/2$: converts X -rotation to Y -rotation

Dispersive Readout

- $\hat{H} \approx (\omega_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\omega_q}{2} \hat{\sigma}_z$
- SNR $\propto \chi/\kappa$; effective Purcell suppression: $\kappa_{\text{eff}}(\omega_q) \ll \kappa(\omega_r)$

5. QUBIT CHARACTERISATION PROTOCOLS

Reference Frames

Region	Axis	Physical meaning
North pole	$+z$	$ 0\rangle$ (ground)
South pole	$-z$	$ 1\rangle$ (excited)
Equator	xy -plane	Superposition
Precession	$\phi(t)$	Rotation at ω_{01}

At resonance ($\omega_r = \omega_{01}$): $H' = 0$ in rotating frame.

T_1 Measurement (Inversion Recovery)

- Sequence: X_π — wait τ — measure
- Decay: $\langle Z \rangle(\tau) \propto e^{-\tau/T_1}$

T_2^* Measurement (Ramsey)

- Sequence: $X_{\pi/2}$ — wait τ — $X_{\pi/2}$ — measure
- Signal: oscillates and decays as $e^{-\tau/T_2^*}$

T_2 Measurement (Hahn Echo)

- Sequence: $X_{\pi/2}$ — wait $\tau/2$ — Y_π — wait $\tau/2$ — $X_{\pi/2}$
- Refocuses static noise: $T_2 \geq T_2^*$

Randomised Benchmarking

- Fit: $p(m) = A + Bp^m$; error per gate $r = (1 - p)/2$
- SPAM fidelity: extract fit at $m = 0$

6. TRANSMISSION LINES

Distributed Parameters

Symbol	Quantity
R_0 [Ω/m]	Resistance per unit length
L_0 [H/m]	Inductance per unit length
C_0 [F/m]	Capacitance per unit length
G_0 [S/m]	Conductance per unit length

Telegrapher Equations

$$\frac{\partial V}{\partial z} = -R_0 i - L_0 \frac{\partial i}{\partial t}, \quad \frac{\partial i}{\partial z} = -G_0 V - C_0 \frac{\partial V}{\partial t}$$

Lossless line ($R_0 = G_0 = 0$): wave equation

$$\frac{\partial^2 V}{\partial z^2} = L_0 C_0 \frac{\partial^2 V}{\partial t^2}$$

Wave Parameters

- Phase velocity: $v = 1/\sqrt{L_0 C_0}$
- Characteristic impedance: $Z_0 = \sqrt{L_0/C_0}$
- General solution: $V(z, t) = V^+(t - z/v) + V^-(t + z/v)$
- Currents: $i^+ = V^+/Z_0$; $i^- = -V^-/Z_0$

Reflections & Matching

- Voltage reflection: $\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$
- Current reflection: $\rho_i = -\Gamma_L$
- Matched load: $R_L = Z_0 \Rightarrow \Gamma_L = 0$ (no reflection)

Power

- Instantaneous: $p = Vi$
- Average (travelling wave): $P_{\text{avg}} = V_{\text{rms}}^2/Z_0 = I_{\text{rms}}^2 Z_0$

7. S-PARAMETERS

Power-Wave Definitions

$$a_i = \frac{V_i + Z_{ri} I_i}{2\sqrt{Z_{ri}}}, \quad b_i = \frac{V_i - Z_{ri} I_i}{2\sqrt{Z_{ri}}}$$

- $|a_i|^2, |b_i|^2$: incident/reflected power [W]
- Net power: $P_i = |a_i|^2 - |b_i|^2$
- Propagation: $a(x) = a(0)e^{-jkx}$; $b(x) = b(0)e^{+jkx}$

S-Matrix (N-port)

$$\mathbf{b} = \mathbf{S} \mathbf{a}, \quad S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_{k \neq j} = 0}$$

Two-port:

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

- Reciprocal: $S_{ij} = S_{ji}$; Lossless: $\mathbf{S} \mathbf{S}^\dagger = \mathbf{I}$

Impedance Discontinuity (Z_1/Z_2 junction)

$$\mathbf{S} = \frac{1}{Z_1 + Z_2} \begin{bmatrix} Z_2 - Z_1 & 2\sqrt{Z_1 Z_2} \\ 2\sqrt{Z_1 Z_2} & Z_1 - Z_2 \end{bmatrix}$$

- $S_{11} = (Z_2 - Z_1)/(Z_1 + Z_2)$; $S_{22} = -S_{11}$
- $S_{21} = S_{12} = 2\sqrt{Z_1 Z_2}/(Z_1 + Z_2)$

VNA Measurement

- RX_{REF} : measures a_1 ; RX_{TEST1} : b_1 ; RX_{TEST2} : b_2
- Forward: $S_{11} = b_1/a_1$, $S_{21} = b_2/a_1$
- Reverse: $S_{22} = b_2/a_2$, $S_{12} = b_1/a_2$

Directional Coupler

$$\text{Coupling [dB]} = -10 \log \left(\frac{P_{\text{fwd,coupled}}}{P_{\text{incident}}} \right)$$

$$\text{Isolation [dB]} = -10 \log \left(\frac{P_{\text{rev,coupled}}}{P_{\text{incident}}} \right)$$

$$\text{Directivity [dB]} = \text{Isolation} - \text{Coupling}$$

8. COMPLEX SIGNALS & MODULATION

Euler / Phasor Identities

$$e^{\pm j\omega t} = \cos \omega t \pm j \sin \omega t$$

$$\cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}, \quad \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

General Sinusoid

$$v(t) = A \sin(2\pi f t + \phi)$$

Amplitude Modulation (AM)

$$s(t) = [A_c + A_m \cos 2\pi f_m t] \cos 2\pi f_c t$$

Using $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$s(t) = A_c \cos 2\pi f_c t + \frac{A_m}{2} \cos 2\pi(f_c \pm f_m)t$$

I/Q (Quadrature) Modulation

- Complex baseband envelope: $g(t) = g_1(t) + jg_2(t)$
- Modulated signal: $v(t) = \text{Re}\{g(t)e^{j2\pi f_c t}\}$

$$= g_1(t) \cos(2\pi f_c t) - g_2(t) \sin(2\pi f_c t)$$

I/Q Demodulation

- Multiply by $e^{-j2\pi f_c t}$, then LPF:

$$z_1(t) = \frac{1}{2}g_1(t), \quad z_2(t) = \frac{1}{2}g_2(t)$$

Magnitude & Phase Spectra

$$|G(f)| = 2\sqrt{A^2(f) + B^2(f)}, \quad \angle G(f) = \arctan\left(\frac{B(f)}{A(f)}\right)$$

9. DIGITAL SIGNAL PROCESSING

Sampling

- Sampling rate: $f_s = 1/t_s$
- Nyquist-Shannon: $f_s > 2(f_h - f_l)$
- Alias frequency: $f_{\text{alias}} = |f - nf_s|$
- DAC image locations: $Nf_s \pm f$

Quantisation

- Step size: $q = V_{\text{FS}}/(2^N - 1)$; ADC resolution: $\Delta = V_{\text{FS}}/2^n$

- Dynamic range: $\text{DR} = 20 \log_{10}(2^N) \approx 6.02N \text{ dB}$
- Quantisation noise power: $P_q = q^2/12$
- SFDR: $P_{\text{fundamental}}/P_{\text{biggest spur}}$

Fixed-Point Format $Q_{n.b}$

- Unsigned: range $[0, 2^n - 2^{-b}]$
- Signed 2's complement: $[-2^{n-1}, 2^{n-1} - 2^{-b}]$
- Resolution: 2^{-b}
- Add/sub: +1 bit for overflow guard
- Multiply: $Q(m_1.n_1) \times Q(m_2.n_2) \rightarrow Q(m_1+m_2.n_1+n_2)$

Multirate Processing

- Decimation by M : $f_d = f_s/M$
- Interpolation by L : $f_u = Lf_s$
- First image after interpolation by K : $f_{\text{image}} = Kf_c - f_0$

Data Converters

- Interleaved ADC phase (n -th of m): $\theta_n = 2\pi(n-1)/m$
- DAC sinc response (normal mode): $H(f) = \frac{\sin x}{x}$, $x = \pi f/f_s$
- DAC mix-mode response: $H(f) = \frac{\sin^2 x}{x}$, $x = \pi f/(2f_s)$

10. SEMICONDUCTOR QUBITS — ELZERMAN READOUT

Energy Hierarchy

$$\Delta E_{\text{orbital}} \gg \Delta E_Z \gg k_B T$$

- Zeeman energy ($B = 10 \text{ T}$): $\Delta E_Z \approx 200 \mu\text{eV}$
- Thermal energy ($T = 0.3 \text{ K}$): $k_B T \approx 25 \mu\text{eV}$
- Orbital spacing: $\Delta E_{\text{orbital}} \approx 1.1 \text{ meV}$

Bias Tuning

$$V_M = V_{M0} + \frac{\Delta E_Z}{2e}$$

Places Fermi energy midway between spin-up and spin-down levels.

Measurement Fidelity

- $\alpha = P(\text{"down"} \mid \uparrow)$; $\beta = P(\text{"up"} \mid \downarrow)$
- Decay/detection error: $\beta_1 = \frac{1}{1 + T_1 \Gamma_{\downarrow}}$
- Combined β : $1 - \beta \approx (1 - \beta_1)(1 - \beta_2) + \alpha\beta_1$
- Spin-up fidelity: $1 - \alpha$
- Visibility: $\mathcal{V} = 1 - \alpha - \beta$

Two-Qubit Exchange Interaction

$$H_0 \sim \sum_i \omega_i \hat{\sigma}_{zi}$$

$$J = f_2 - f_1 \quad \text{or} \quad J = f_4 - f_3$$

where $f_{1,2}$ are resonance frequencies of q_0 conditioned on state of q_1 , and vice-versa.